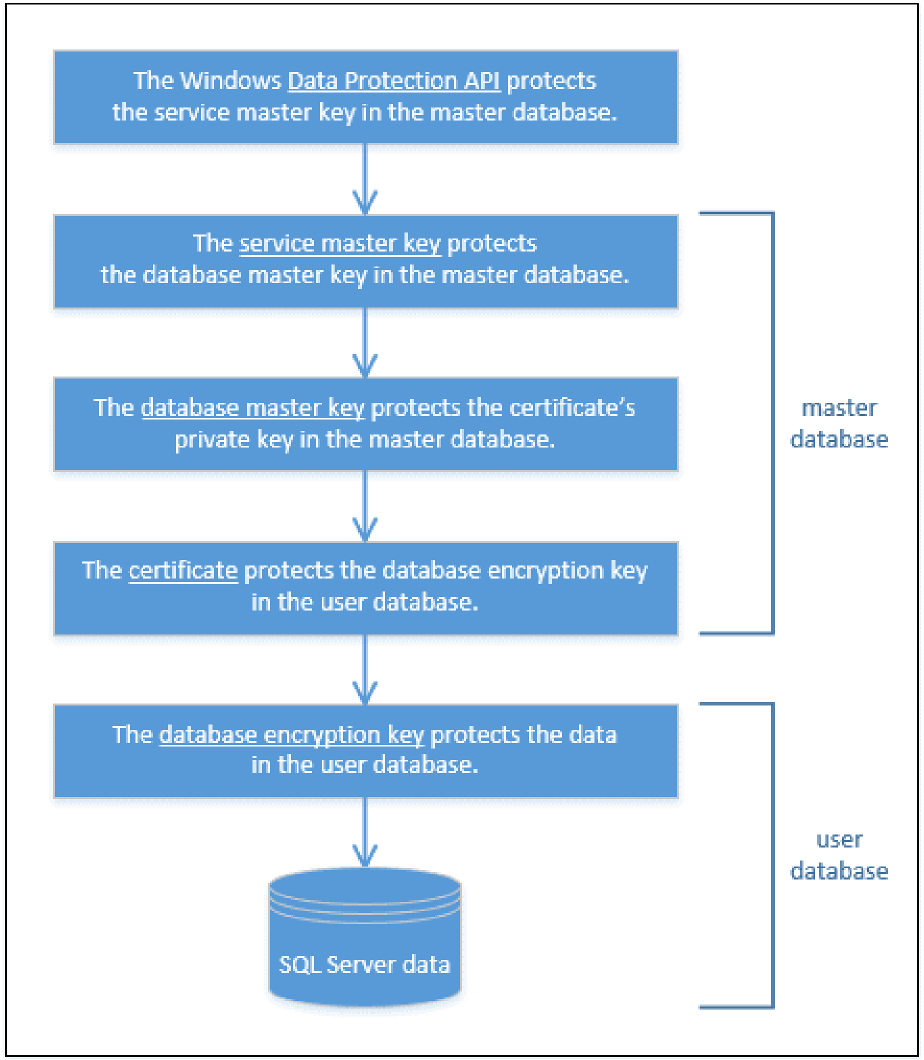
TDE Encryption and Its Implementation

Transparent Data Encryption (TDE)

- With the release of SQL Server 2008, Microsoft expanded the database engine’s security capabilities by adding Transparent Data Encryption (TDE), a built-in feature for encrypting data at rest.
- TDE protects the physical media that hold the data associated with a user database, including the data and log files and any backups or snapshots.
- Encrypting data at rest can help prevent those with malicious intent from being able to read the data should they manage to access the files.
- SQL Server TDE takes an all-or-nothing approach to protecting data. When enabled, TDE encrypts all data in the database.
- You cannot pick-and-choose like you can with column-level encryption.
- TDE is relatively easy to enable.
- TDE requires planning but can be implemented without changing the database.

TDE Encryption Hierarchy

- Starting with the Windows DPAPI at the top and the SQL Server data at the bottom.
- the next level down our hierarchy is a certificate, which you also create in the master database.
- The DMK protects the certificate, and the certificate protects the database encryption key (DEK) in the user database.
- The DEK is specific to TDE and is used to encrypt the data in the user database in which the key resides.
- we can deduce that to implement TDE on a user database, we must take the following steps:
    - Create the DMK in the master database, if it doesn't already exist.
    - Create a certificate in the master database for securing the DEK.
    - Create the DEK in the user database to be encrypted.
    - Enable TDE on the user database.



TDE Keys Management

- Importance of backing up the SMK, DMK, and certificate.
- If anything goes wrong in our production environment or we need to restore or move an encrypted database, we might need those keys or certificate, so we better make sure we have copies of them, stored securely in a separate location.
- You should back up your certificates and keys, preferably right after you create them. This is also true for the SMK, before you start relying on it to protect your DMKs.